

Due dates 31 July 2026
Date functions Exercise 5

Extract Date Parts

Question 1

```
SELECT order-id,  
       customer-id,  
       Order-date,  
       DAYNAME(Order-date) AS day-name  
FROM orders;
```

Results: orders

order-id	customer-id	Order-date	day-name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

```
② SELECT Customer-id,  
       Customer-name,  
       Signup-date,  
       MONTHNAME(signup-date) AS signup-month-name  
FROM Customer-signups;
```

Results: Customer-signups

Customer-id	Customer name	Signup-date	Signup-month-name
201	John	2026-01-15	January
202	May	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

Q3 SELECT sale-id,

product-name,

sale-date,

MONTH(sale-date) AS month, sale-month

FROM sales;

Results: Sales

sale-id	product-name	sale-date	sale-month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

Q4 SELECT transaction-ids

customer-id,

transaction-date

YEAR(transaction-date) AS transaction-year

FROM transactions;

Results: Transactions

transaction-id	customer-id	transaction-date	transaction-year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

Q5 SELECT delivery-id,

customer-id,

delivery-date,

DAY(delivery-date) AS day-of-month

FROM deliveries;

Results: deliveries

delivery-id	customer-id	delivery-date	day-of-month
D001	101	2026-05-01	1
D002	102	2026-05-02	2
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

(Question 2 (Current Date, Convert and Format))

Q1 SELECT employee-id,

employee-name,

department,

CURRENT-DATE AS today-date

FROM employees;

Results: employees

employee-id	employee-name	department	today-date
301	Mandi	Sales	2026-07-30
302	Brian	IT	2026-07-30
303	Lerato	Finance	2026-07-30
304	Sipha	HR	2026-07-30
305	Aisha	Marketing	2026-07-30

② SELECT order-id,

customer-id,

order-date-text,

TO-DATE(order-date-text, 'YYYY-MM-DD')

AS order-date

FROM online-orders;

Results: online-orders

Order-id	customer-id	order-date-text	order-date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

③ SELECT payment-id,

customer-id,

payment-date

TO-CHAR(payment-date, 'YYYY-MM-DD')

AS formatted-payment-date

FROM payment-dates;

Results: payment-dates

Payment-id	customer-id	Payment-date	formatted-payment-date
P001	101	2026-01-05	
P002	102	2026-02-10	
P003	103	2026-03-15	
P004	104	2026-04-20	
P005	105	2026-05-25	

④ SELECT customer-id,

customer-name,

last-purchase-date,

DATEDIFF(CURRENT_DATE, last-purchase-date)

AS days-since-last-purchase

FROM customers-purchases;

Results: customers-purchases

Customer-id	Customer-name	last-purchase-date	days-since-last-purchase
501	John	2026-05-01	90
502	Mary	2026-05-10	81
503	Peter	2026-05-15	76
504	Sarah	2026-05-20	71
505	Thabo	2026-05-25	66

⑤ SELECT order-id,

customer-id,

order-date

DATEADD(DAY, 7, Order-date) AS expected-

delivery-date

FROM shipping-orders

Results: shipping-orders

Order-id	customer-id	order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Question 3: Multi-Extract and Date Filtering

Q1 SELECT booking-id,

customer-id,

booking-date,

YEAR(booking-date) AS booking-years,

MONTH(booking-date) AS booking-months

DAY(booking-date) AS booking-day

FROM bookings;

Results: bookings

booking-id	customer-id	booking-date	booking-year	booking-month	booking-day
B001	101	2026-01-12	2026	January 1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	09
B005	105	2026-05-27	2026	5	27

Q2 SELECT order-id,

customer-id,

order-date,

YEAR(order-date) AS order-years,

MONTH(order-date)

WHERE YEAR(order-date) = 2026;

Results: yearly-orders

order-id	customer-id	order-date	order-year	order-month	order-day
O001	101				
O002	102				
O003	103				
O004	104				
O005	105				

Results: yearly-orders

order-id	customer-id	order-date	order-year	order-month	order-day
O004	104	2026-05-25	2026	5	25
O005	105	2026-05-25	2026	5	25

Q3 SELECT order-id,

customer-id,

order-date,

MONTH(order-date) AS order-month,

amount

FROM monthly-orders

WHERE MONTH(order-date) = 3;

Results: monthly-orders

order-id	customer-id	order-date	order-month	amount
O003	103	2026-03-10	3	800
O004	104	2026-03-25	3	1500

Q4 SELECT subscription-id,

customer-id,

start-date,

LASTDAY(start-date) AS month-end-date

FROM subscriptions;

Results: subscriptions

subscription-id	customer-id	start-date	month-end-date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

Q5 SELECT send_id,

customer_id,

send_date,

FIRST DATE(send_date) AS month_start_date

FROM campaign_send;

Results: campaign_send

Send_id	customer_id	send_date	Month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-01	2026-04-01
C005	105	2026-05-21	2026-05-01

Questions 16-20 Advanced Date work

Q6 SELECT invoice_id,

customer_id,

invoice_date,

TO_CHAR(invoice_date, 'Month YYYY') AS

invoice_month_year

FROM invoice_dates;

Results:

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

Q7 SELECT customer_id,

customer_name,

date_of_birth,

DATEDIFF(2026, date_of_birth, CURRENT_DATE)

AS customer_age

FROM customer_birthdays

Results

customer_id	customer_name	date_of_birth	customer_age
901	John	1978-05-10	28
902	Mary	1990-02-20	35
903	Peter	1982-03-15	24
904	Sarah	1975-12-01	40
905	Thomas	2000-07-30	26

Q8 SELECT order_id,

customer_id,

order_date,

DRAWNAME(customer) AS day_name,

WHEN DATEDIFF(order_date, TRUNC(order_date, 'WEEKEND'))

'weekend'

ELSE 'weekend'

END AS day_type

FROM weekend_orders;

Results

order_id	customer_id	order_date	day_name	day_type
9001	101	2026-05-01	Friday	Weekend
9002	102	2026-05-02	Saturday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekday
9005	105	2026-05-05	Tuesday	Weekday

(Q4) SELECT transaction_id,
 customer_id,
 transaction_date,
 @ QUARTER(transaction_date) AS transaction_quarter,
 amount

FROM quarterly_transactions;

Results

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

(Q5) SELECT order_id,
 customer_id,
 order_date,
 DATEDIFF(CURRENT_DATE(), order_date) AS
 days_since_order,
 amount

FROM recent_orders

(WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30;

Results

order_id	customer_id	order_date	days_since_order	amount
R001	101	2026-04-01	121	500
R002	102	2026-04-15	107	1200
R003	103	2026-05-01	91	800
R004	104	2026-05-10	82	1500
R005	105	2026-05-25	67	2000